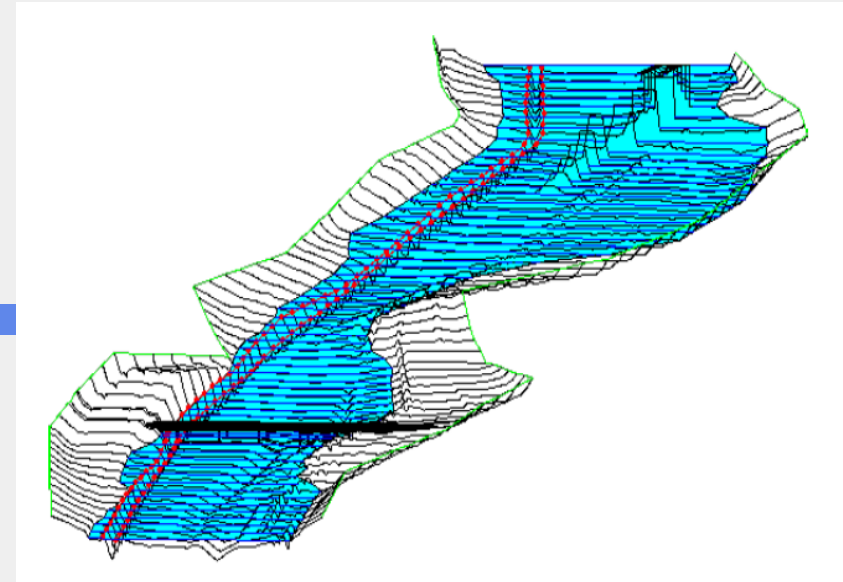


Introduction to HEC-RAS



HEC-RAS

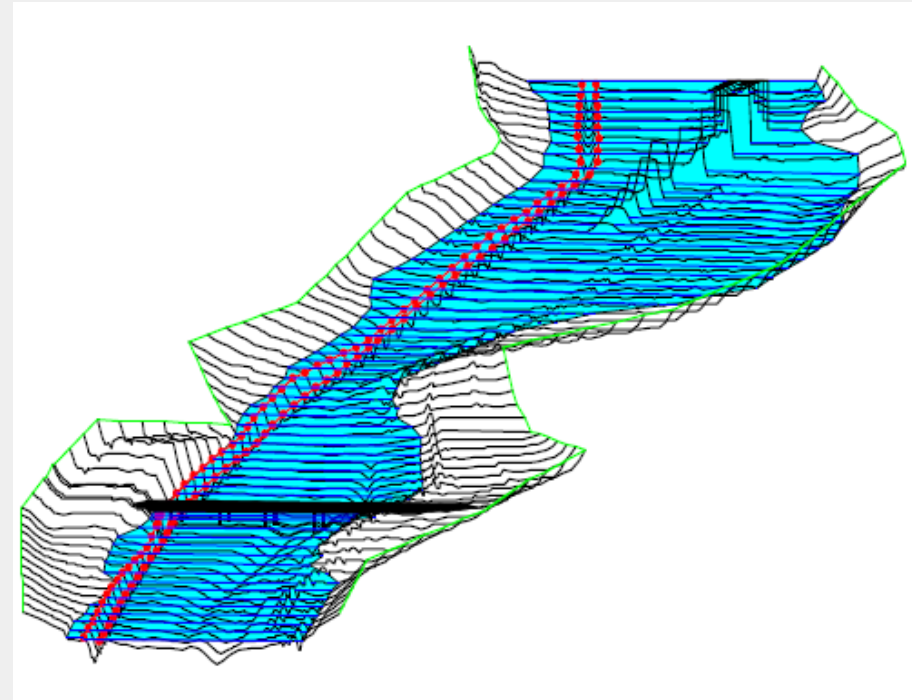
Hydrologic **E**ngineering **C**enter – **R**iver
Analysis **S**oftware

By the U.S. Army Corps of Engineers

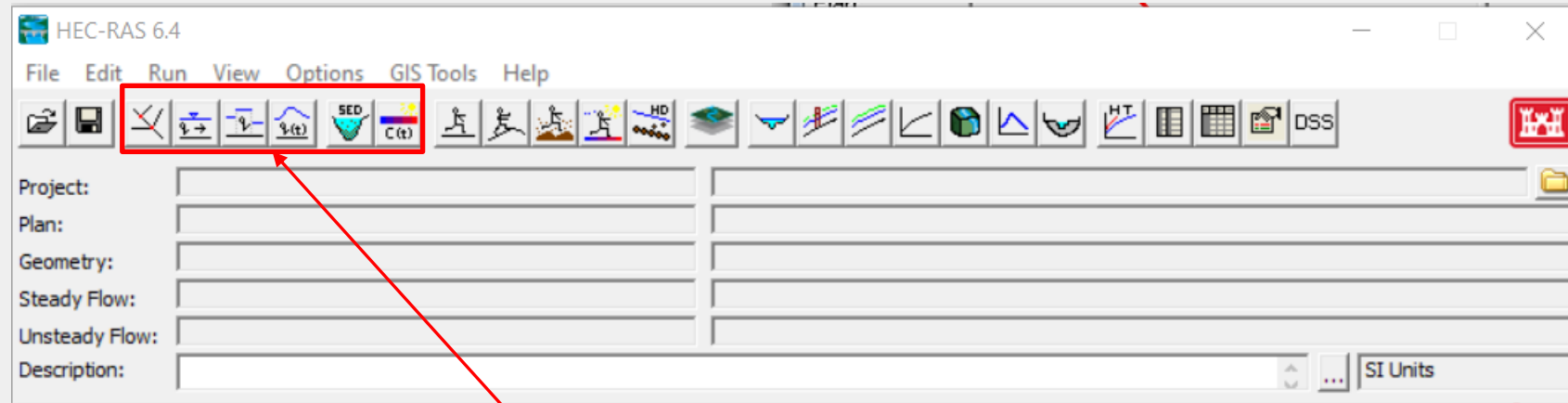
- Software for 1D and 2D hydraulic modelling of channels, rivers and their floodplains.

- Freely available

<http://www.hec.usace.army.mil/software/hec-ras/>



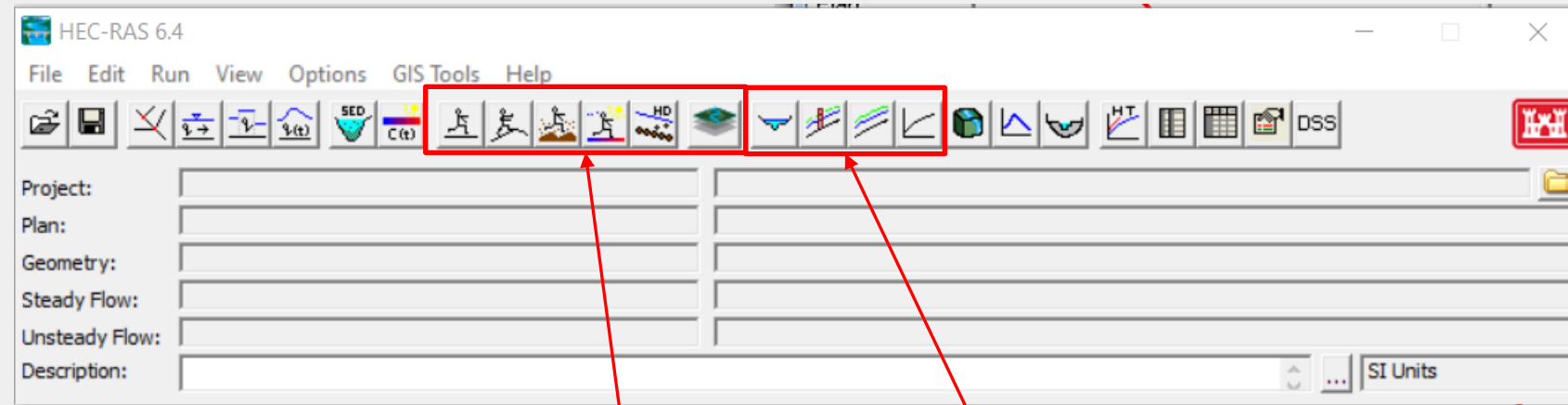
HEC-RAS



Required for each simulation:

- Geometry data
- Flow data (steady or unsteady)
- Potentially sediment data
- Potentially water quality data

HEC-RAS



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- Geometry data
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- Potentially sediment data
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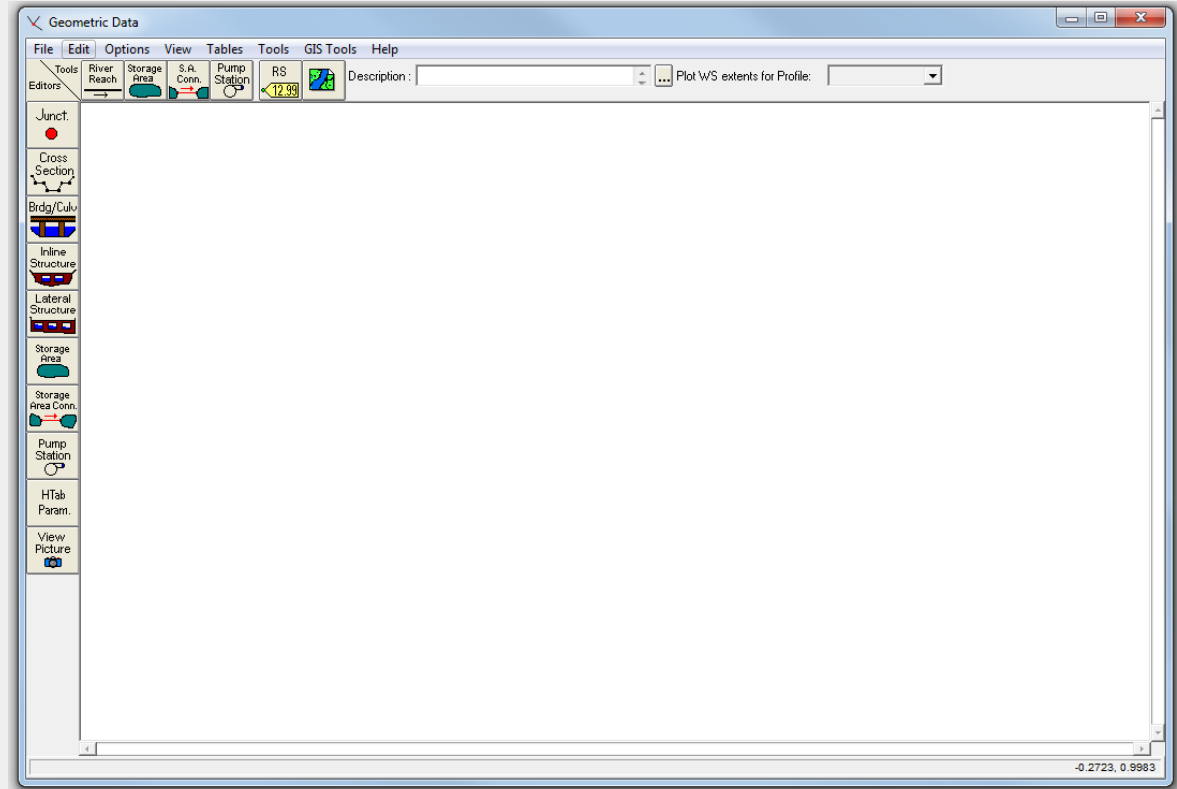
Simulation
settings & start

Results

Channel geometry

Input of geometry data in 2 stages:

1. Channel network
(Tributary, bridges, etc.)
2. Cross sections

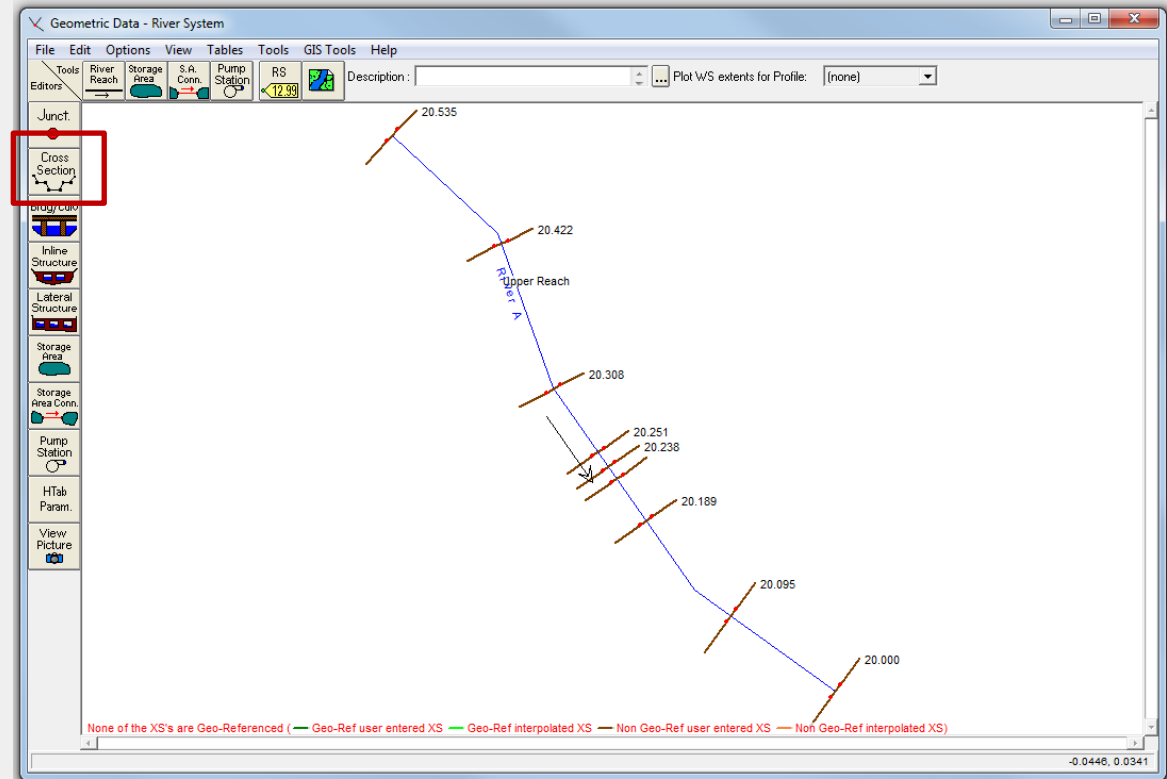


Channel network



Input of geometry data in 2 stages:

1. **Channel network**
(Tributary, bridges, etc.)
2. Cross sections



Cross sections



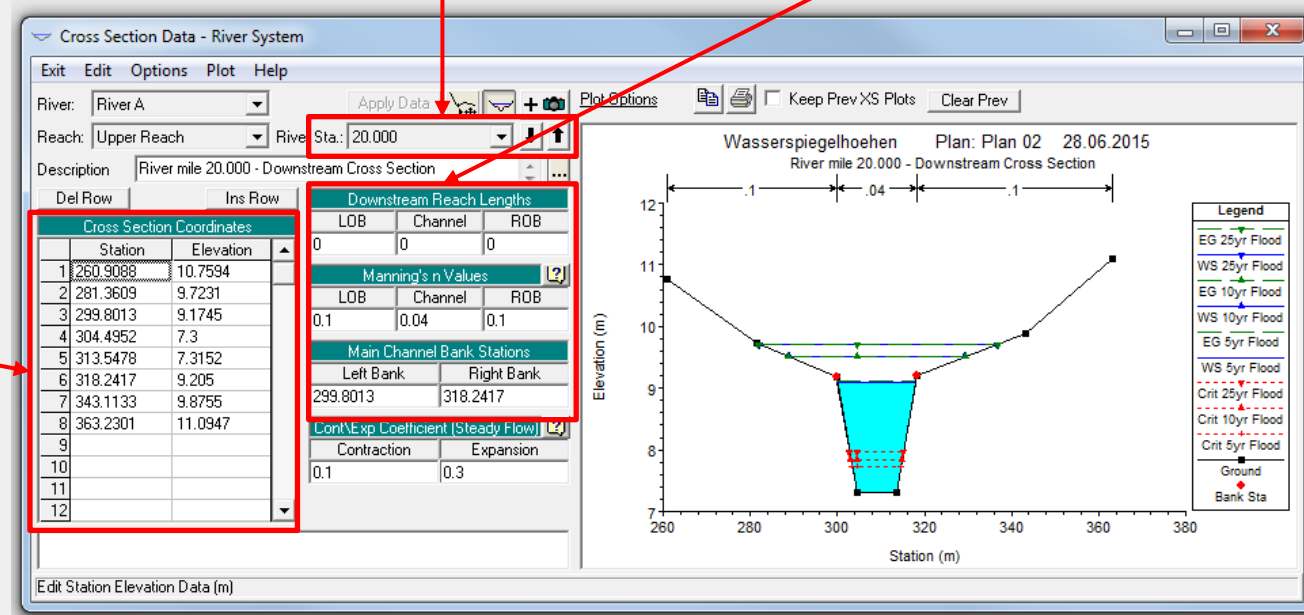
Input of geometry data in 2 stages:

1. Channel network
(Tributary, bridges, etc.)
2. **Cross sections**

Profile number

Distance to the end of
the channel, roughness,
main channel

Geometry

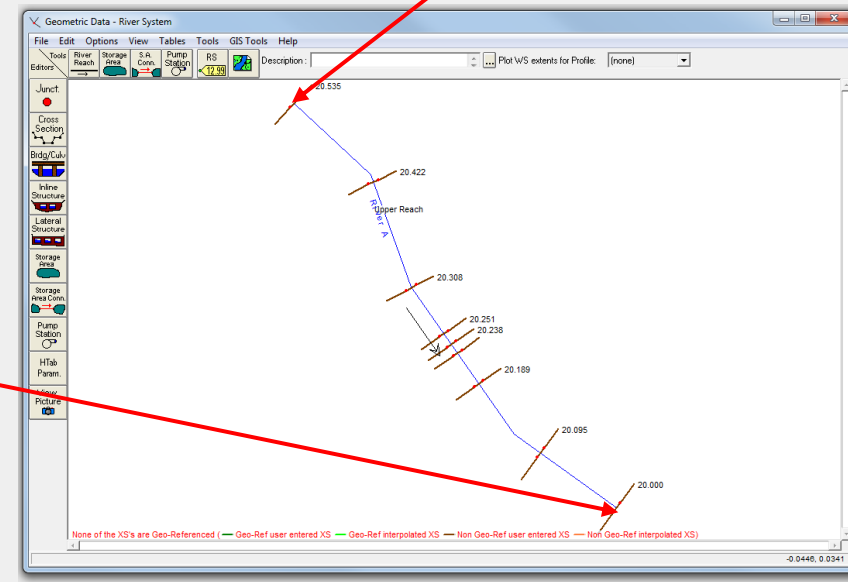
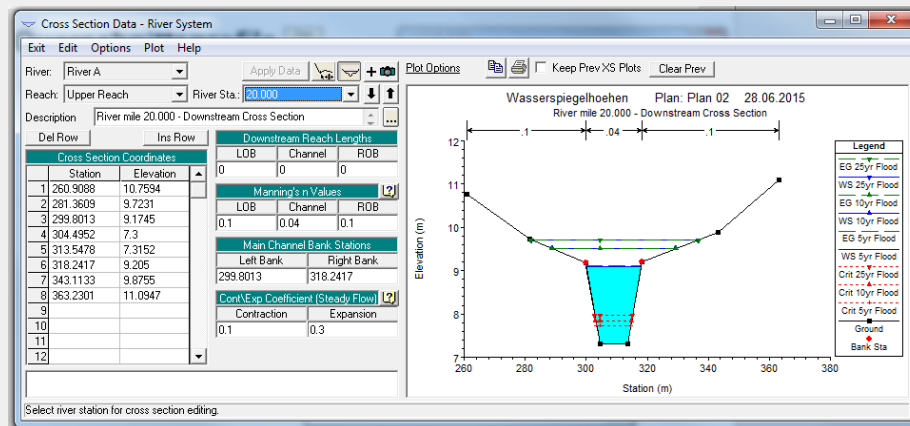
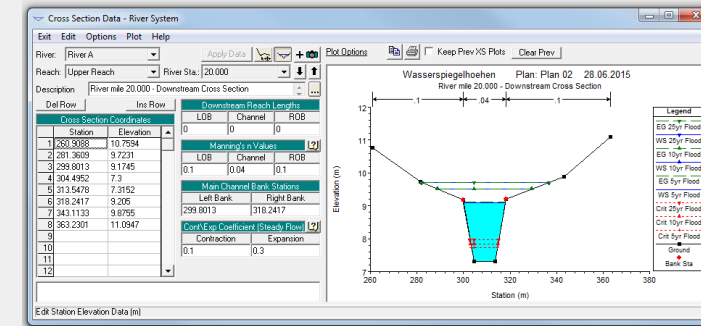


Cross sections



Minimum of two:

- Upper and lower boundary of the channel
- Profiles in between added as available
- or interpolate

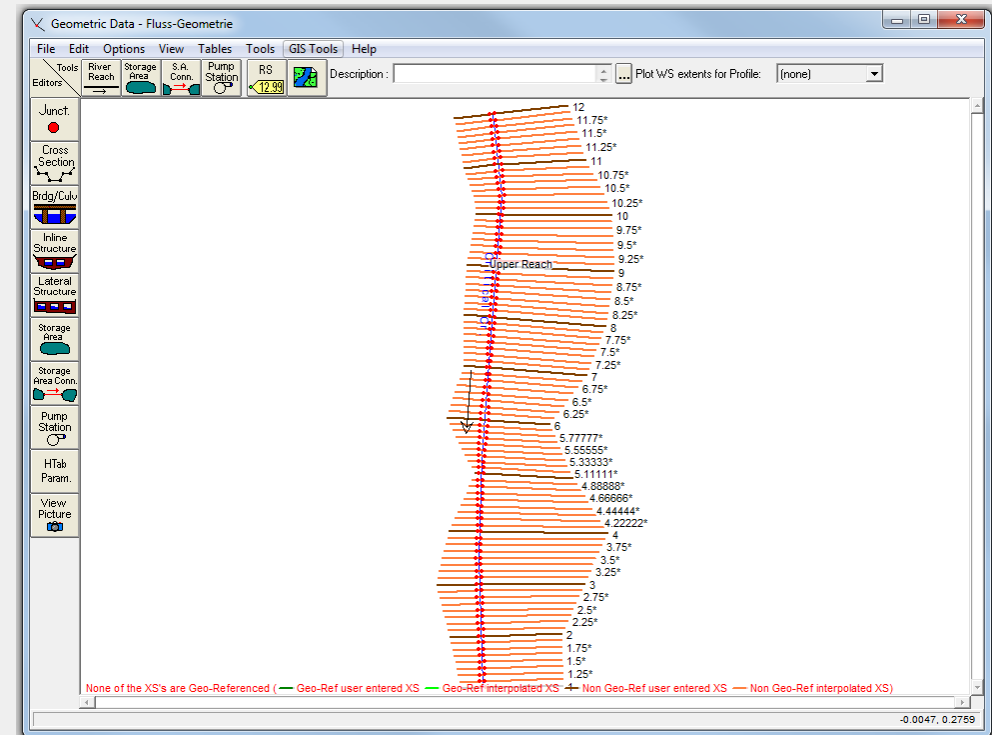


Cross sections



Minimum of two:

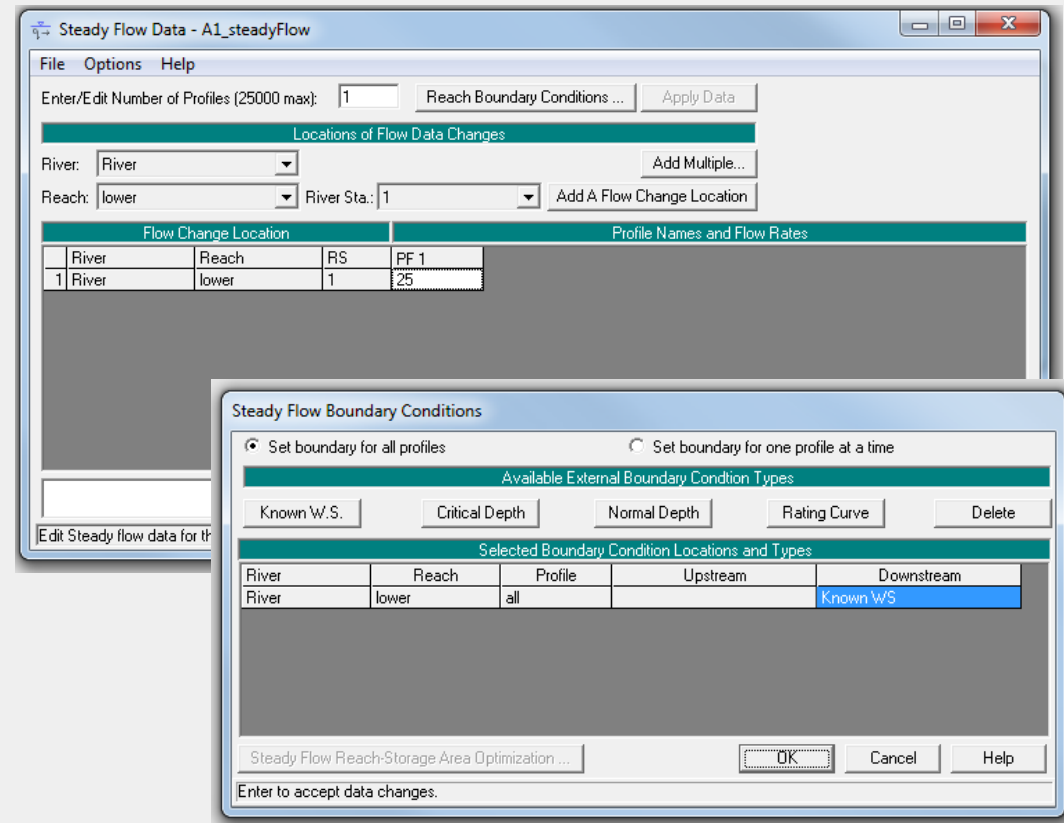
- Upper and lower boundary of the channel
- Profiles in between added as available
- or interpolate



Steady flow data

Input of steady flow data

- Discharge [m^3/s]
- Boundary conditions (BCs)
 - No change in flow regime: 1 BC
 - Change in flow regime: 2 BCs



The image shows two overlapping dialog boxes from the HEC-RAS software. The background dialog is titled "Steady Flow Data - A1_steadyFlow" and contains fields for "Enter/Edit Number of Profiles (25000 max):" set to 1, and buttons for "Reach Boundary Conditions ..." and "Apply Data". It also has dropdowns for "River:" (set to "River") and "Reach:" (set to "lower"), with a "River Sta.:" field set to 1. Below these are two tables. The first table, "Locations of Flow Data Changes", has columns "River", "Reach", "RS", and "PF 1", with one row showing "River", "lower", "1", and "25". The second table, "Profile Names and Flow Rates", is empty. The foreground dialog is titled "Steady Flow Boundary Conditions" and has two radio buttons: "Set boundary for all profiles" (selected) and "Set boundary for one profile at a time". It features buttons for "Known W.S.", "Critical Depth", "Normal Depth", "Rating Curve", and "Delete". Below these is a table titled "Selected Boundary Condition Locations and Types" with columns "River", "Reach", "Profile", "Upstream", and "Downstream". The first row shows "River", "lower", "all", and "Known WS" in the "Downstream" column. At the bottom of the foreground dialog are buttons for "Steady Flow Reach-Storage Area Optimization ...", "OK", "Cancel", and "Help", along with the text "Enter to accept data changes."

Steady Flow Data - A1_steadyFlow

File Options Help

Enter/Edit Number of Profiles (25000 max): 1 Reach Boundary Conditions ... Apply Data

Locations of Flow Data Changes

River: River Add Multiple...

Reach: lower River Sta.: 1 Add A Flow Change Location

Flow Change Location			Profile Names and Flow Rates	
River	Reach	RS	PF 1	
1 River	lower	1	25	

Edit Steady flow data for the

Steady Flow Boundary Conditions

☒ Set boundary for all profiles ☐ Set boundary for one profile at a time

Available External Boundary Condition Types

Known W.S. Critical Depth Normal Depth Rating Curve Delete

Selected Boundary Condition Locations and Types				
River	Reach	Profile	Upstream	Downstream
River	lower	all		Known WS

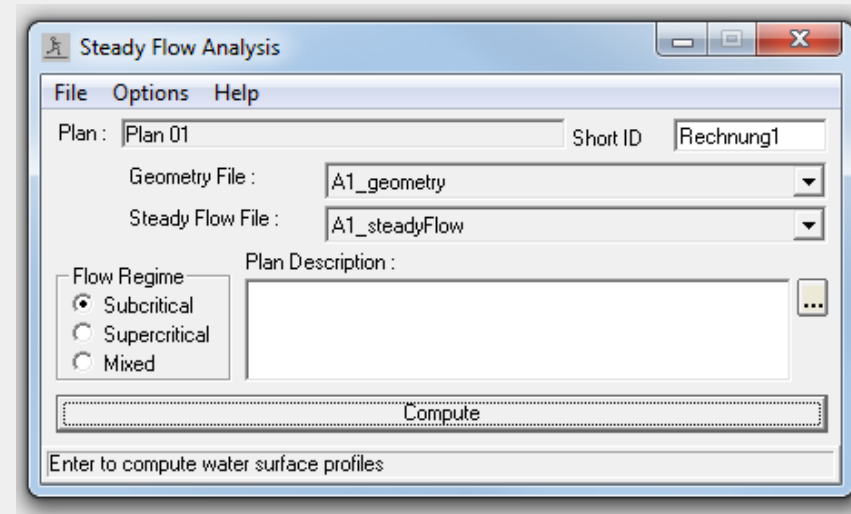
Steady Flow Reach-Storage Area Optimization ... OK Cancel Help

Enter to accept data changes.

Steady simulation

Running the simulation,
Calculation of the water level

- Input regarding sub- or supercritical flow, or mixed.
- Input can be checked upon completion of the calculations.



Computer Lab

Calculation of the water surface profile upstream of a dam

- Constant cross section
- Trapezoidal cross section
- Steady flow

